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Optimization



Bill Howe

You will learn how to optimize a cost function using gradient descent, including popular variants that use randomization and parallelization to improve performance. You will gain an

Lesson 11: Gradient Descent



Optimization by Gradient Descent 3 min

(/learn/predictive-analytics/lecture/68oAE/optimization-by-

Help Centre



gradient-descent)

 Gradient Descent
Visually 4 min

(/learn/predictive-
analytics/lecture/qOSum/gradient-descent-
visually)

 Gradient Descent in
Detail 3 min

(/learn/predictive-
analytics/lecture/xoaXa/gradient-descent-in-
detail)

 Gradient Descent:
Questions to
Consider 4 min

(/learn/predictive-
analytics/lecture/edAcy/gradient-descent-
questions-to-consider)

Lesson 12: Generalizing the Cost Function

 Intuition for Logistic
Regression 4 min

(/learn/predictive-
analytics/lecture/FecmG/intuition-for-logistic-
regression)

 Intuition for Support
Vector Machines 4 min

(/learn/predictive-
analytics/lecture/6hPg3/intuition-for-support-
vector-machines)

 Support Vector Machine
Example 3 min

(/learn/predictive-
analytics/lecture/2Qh1o/support-vector-
machine-example)

 Intuition for



Regularization 3 min

(/learn/predictive-analytics/lecture/j04O0/intuition-for-regularization)



Intuition for LASSO and Ridge Regression 4 min

(/learn/predictive-analytics/lecture/OdcI5/intuition-for-lasso-and-ridge-regression)

Lesson 13: Algorithmic Considerations



Stochastic and Batched Gradient Descent 5 min

(/learn/predictive-analytics/lecture/eCynR/stochastic-and-batched-gradient-descent)



Parallelizing Gradient Descent 4 min

(/learn/predictive-analytics/lecture/RhkFB/parallelizing-gradient-descent)